Savitribai Phule Pune University
S.Y.B.Sc(Computer Science) NEP - 2023
S.Y.B.Sc.(CS) Practical Examination MAR/APRIL – 2025 - 2026
Lab Course SEC-251-CS-P
Computer Networks

Duration: 3 Hours **Maximum Marks: 35**

Q1) A Write the answer of the following question (ANY 5) [10 Marks]

a) What is the purpose of a modem in a network? Can we connect directly
to the internet without it?

ANSWER

A modem (Modulator–Demodulator) converts digital signals from a computer into analog signals for transmission over telephone or cable lines and converts incoming analog signals back to digital.

It connects the local network to the Internet Service Provider (ISP).
Without a modem (or modem-built-in device like a router/ONT), a computer cannot directly access the internet through ISP lines.

b) What is a MAC address and how is it different from an IP address?

ANSWER

MAC Address (Media Access Control Address):

A unique physical address assigned to a network interface card (NIC).

Example: 00:1A:2B:3C:4D:5E

Works at the Data Link Layer.

IP Address (Internet Protocol Address):

A logical address assigned to identify a device on a network.

Example: 192.168.1.1

Works at the Network Layer.

Difference:
MAC address is permanent hardware identification, while IP address is logical and can change depending on the network.

d) Command to assign a static IP in Linux

ANSWER

Using the ip command:

sudo ip addr add 192.168.1.10/24 dev eth0

Explanation:

ip addr add → Adds an IP address to a network interface

192.168.1.10/24 → Static IP address with subnet mask

dev eth0 → Network interface name

c) What is a loopback IP address? What is its range?

ANSWER

A loopback IP address is used by a computer to communicate with itself for testing network software and configurations.

Common loopback address: 127.0.0.1

Range: 127.0.0.0 – 127.255.255.255

f) Which online tool used to check password strength

ANSWER

Some common password strength checking tools are:

Kaspersky Password Checker

NordPass Password Strength Checker

How Secure Is My Password

These tools analyze password complexity and estimate how long it would take to crack.

Q2) B Write a C program to check password is strong or weak.

[20marks]

ANSWER

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

int main() {
    char password[100];
    int i;
    int hasUpper = 0, hasLower = 0, hasDigit = 0, hasSpecial = 0;

    printf("Enter password: ");
    scanf("%s", password);

    if(strlen(password) < 8) {
        printf("Password is Weak (Minimum length should be 8 characters)\n");
        return 0;
    }

    for(i = 0; i < strlen(password); i++) {
        if(isupper(password[i]))
            hasUpper = 1;
        else if(islower(password[i]))
            hasLower = 1;
        else if(isdigit(password[i]))
            hasDigit = 1;
        else
            hasSpecial = 1;
    }

    if(hasUpper && hasLower && hasDigit && hasSpecial)
        printf("Password is Strong\n");
    else
        printf("Password is Weak\n");

    return 0;
}
```

Q3. Viva [5 Marks]

Savitribai Phule Pune University
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Q1) A Write the answer of the following question (ANY 5) [10Marks]

a) What is the functional difference between a switch and a router?

ANSWER

A switch connects multiple devices within the same local network (LAN) and forwards data using MAC addresses.

A router connects different networks and forwards data packets using IP addresses.

b) How many hosts are possible in a /26 subnet?

ANSWER

A /26 subnet means 26 bits for network and 6 bits for hosts.

Total addresses = 2^6=64

Usable hosts = 64 − 2 = 62

Answer: 62 hosts

c) What are the differences between Cat5e, Cat6, and Cat7 Ethernet cables?

ANSWER

Cable Type	**Speed**	**Bandwidth**
Cat5e	**Up to 1 Gbps**	**100 MHz**
Cat6	**Up to 10 Gbps (short distance)**	**250 MHz**
Cat7	**Up to 10 Gbps**	**600 MHz**

Cat6 and Cat7 provide better shielding and lower interference than Cat5e.

f) What are the two analysis methods used for analyzing malware?

ANSWER

Static Analysis – Examining the malware code without executing it.

Dynamic Analysis – Running the malware in a controlled environment to observe its behavior.

g) What is the use of the ping command in networking?

The ping command is used to test connectivity between two network devices.
It sends ICMP echo request packets to check whether a host is reachable and measures response time (latency).

Example:

ping google.com

Q2) B Write a C program to implement framing using character count method in the data link.

[20Marks]

ANSWER

```
#include <stdio.h>
#include <string.h>

int main() {
   char data[100];
   int count;

   printf("Enter data: ");
   scanf("%s", data);

   count = strlen(data) + 1;   // +1 for count field

   printf("\nFrame using Character Count Method:\n");
   printf("%d%s\n", count, data);

   return 0;
}
```

Q3. Viva

[5 Marks]

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Q1) A Write the answer of the following question (ANY 5) [10Marks]

a) What does the command tracert or traceroute do?

ANSWER

The tracert (Windows) or traceroute (Linux) command is used to trace the path that data packets take from the source computer to a destination host.
It shows all the routers (hops) along the route and the time taken for each hop.

Example:

traceroute google.com

c) What does nslookup do?

ANSWER

The nslookup (Name Server Lookup) command is used to query DNS servers to obtain domain name or IP address information.

Example:

nslookup google.com

It returns the IP address of the domain.

d) List two advantages and two disadvantages of dynamic IP configuration

ANSWER

Advantages:

Easy to manage using DHCP server.

Reduces manual configuration errors.

Disadvantages:

IP address may change frequently.

Requires a DHCP server to function properly.

f) What is the function of a switch in a computer network?

ANSWER

A switch connects multiple devices in a Local Area Network (LAN) and forwards data frames to the correct device using MAC addresses.
It improves network efficiency and reduces collisions.

g) What is network topology? What are its applications?

ANSWER

Network topology is the physical or logical arrangement of devices and connections in a computer network.

Applications:

Used in LAN design in offices and schools.

Used in data centers and communication networks for efficient data transmission.

Q2) B Write a C program to display Bus Topology

[20Marks]

```
#include <stdio.h>

int main() {
   int n, i;

   printf("Enter number of nodes: ");
   scanf("%d", &n);

   printf("\nBus Topology Representation:\n\n");

   for(i = 1; i <= n; i++) {
      printf("Node%d", i);
      if(i < n) {
         printf(" --- ");
      }
   }
```

```
    printf("\n\n(All nodes connected to a single backbone cable)");

    return 0;
}
```

Q3. Viva [5 Marks]

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Q1) A Write the answer of the following question (ANY 5) [10Marks]

b) How can you reset or release an IP address obtained from DHCP?

ANSWER

To reset or release an IP address assigned by DHCP, use the following commands:

In Windows (Command Prompt):

Release the current IP address

ipconfig /release

Obtain a new IP address

ipconfig /renew

In Linux:

Release IP:

dhclient -r

Renew IP:

dhclient

c) Explain Supernetting with an example combining IPv4 Class C networks.

ANSWER

Supernetting is the process of combining multiple small networks into one larger network to reduce routing table entries.

Example:

Four Class C networks:

192.168.0.0/24

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

These can be combined into one supernet:

192.168.0.0/22

Explanation:

/24 networks have mask 255.255.255.0

Combining 4 networks reduces mask to /22 (255.255.252.0)

Range covered:

192.168.0.0 – 192.168.3.255

This reduces routing entries and improves efficiency.

d) How do you crack passwords with John the Ripper?

ANSWER

John the Ripper is a password-cracking tool used to test password strength.

Basic steps:

Install John the Ripper

Prepare the password hash file
Example file: password.txt

Run John to start cracking

john password.txt

Use a wordlist attack

john --wordlist=wordlist.txt password.txt

Show cracked passwords

john --show password.txt

John tries dictionary words and brute force methods to find matching passwords.

f) Explain the use of copy running-config startup-config.

ANSWER

This command is used in Cisco routers and switches.

running-config → Current configuration stored in RAM

startup-config → Saved configuration stored in NVRAM

Command:

copy running-config startup-config

Purpose:

Saves the current running configuration permanently.

Ensures the configuration remains after the device restarts.

Without saving, changes will be lost after reboot.

g) What is a Loopback IP address? What is its range?

ANSWER

A Loopback IP address is used by a computer to communicate with itself.
It is mainly used for testing network applications and TCP/IP stack.

Common example:

127.0.0.1 (localhost)

Range of loopback addresses:

127.0.0.0 to 127.255.255.255

Network: 127.0.0.0/8

When you ping 127.0.0.1, the system checks its own network interface without sending traffic to the network.

Q2) B Write a C program that checks private IP ranges.

[20Marks]

ANSWER

```
#include <stdio.h>
#include <stdlib.h>

int main() {
   int a, b, c, d;

   printf("Enter an IP address (format: x.x.x.x): ");
   if(scanf("%d.%d.%d.%d", &a, &b, &c, &d) != 4) {
      printf("Invalid IP format!\n");
      return 1;
   }

   if(a == 10) {
      printf("This is a private IP (Class A range).\n");
   }
   else if(a == 172 && b >= 16 && b <= 31) {
      printf("This is a private IP (Class B range).\n");
   }
   else if(a == 192 && b == 168) {
      printf("This is a private IP (Class C range).\n");
   }
   else {
      printf("This is a public IP.\n");
   }

   return 0;
}
```

Q3. Viva

[5Marks]

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Computer Networks

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Q1) A Write the answer of the following question (ANY 5) [10Marks]

a) What does the ip link set eth0 up command do?

ANSWER

The command:

ip link set eth0 up

Activates the network interface eth0.

Makes it operational so it can send and receive packets.

Equivalent to enabling the interface (turning it "on") without assigning an IP address.

d) What is the purpose of assigning a hostname to a switch?

ANSWER

Identifies the switch in a network.

Makes it easier to manage and troubleshoot.

Appears in the CLI prompt, logs, and network monitoring tools.

Example:

Switch(config)# hostname CoreSwitch
CoreSwitch(config)#

b) How to detect hidden Wi-Fi networks

ANSWER

Hidden Wi-Fi networks do not broadcast their SSID, but they can still be detected using wireless scanning tools.

Steps:

Use a Wi-Fi scanning tool in Linux.

Scan nearby wireless networks.

Hidden networks will appear without an SSID or marked as Hidden.

Example command:

sudo iwlist wlan0 scanning

Explanation:

iwlist → Wireless scanning command

wlan0 → Wireless network interface

scanning → Displays all nearby Wi-Fi networks including hidden ones

f) What is ethical hacking?

ANSWER

Ethical hacking is the practice of intentionally probing and testing computer systems, networks, or applications for vulnerabilities, with permission, to improve security.

Also called white-hat hacking.

Unlike malicious hacking, ethical hackers follow legal and ethical guidelines.

g) What does the show running-config command display?

ANSWER

Displays the current active configuration of a network device (router or switch).

Includes:

Interfaces and IP addresses

Routing protocols

VLANs and port configurations

Access control lists (ACLs)

Hostname and passwords (if configured)

Changes in the running-config are temporary and lost after a reboot unless saved with copy running-config startup-config.

Q2).B Write a C program to display a Ring Topology.

[20Marks]

ANSWER

```
#include <stdio.h>

int main() {
   int nodes, i;

   printf("Enter the number of nodes in the Ring Topology: ");
   scanf("%d", &nodes);

   if(nodes < 3) {
      printf("A ring topology requires at least 3 nodes.\n");
      return 1;
   }

   printf("\nRing Topology Diagram:\n\n");

   // Display nodes in a ring style
   for(i = 1; i <= nodes; i++) {
      printf("Node%d", i);
      if(i != nodes)
         printf(" -> ");
      else
         printf(" -> Node1");  // Loop back to the first node
   }

   printf("\n");

   return 0;
}
```

Q3. Viva

[5Marks]

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Q1) A Write the answer of the following question (ANY 5) [10Marks]

b) List different types of network cables and their uses

ANSWER

Cable Type	Description & Use
Coaxial Cable	**Single copper conductor with insulation & shielding. Used for old LANs, cable TV.**
Twisted Pair (UTP)	**Pairs of copper wires twisted to reduce interference. Used in Ethernet LANs.**
Twisted Pair (STP)	**Same as UTP but with shielding to reduce EMI (Electromagnetic Interference).**
Fiber Optic Cable	**Glass or plastic strands transmitting light. Used for high-speed, long-distance networking.**

c) What is the role of no shutdown in router configuration?

ANSWER

By default, router interfaces are administratively down.

no shutdown command activates the interface, enabling it to send and receive packets.

Example:

```
Router(config)# interface GigabitEthernet0/0
Router(config-if)# no shutdown
```

d) Compare the output of ip addr and ifconfig

ANSWER

Feature	ip addr	ifconfig
Modern / Legacy	**Modern utility (Linux iproute2)**	**Older legacy tool**
Information displayed	**Shows IP addresses, link info, flags**	**Shows IP, netmask, broadcast, MAC**

Flexibility	**Supports multiple advanced operations (routes, links)**	**Basic interface info**
Example command	**ip addr show**	**ifconfig**

g) How many hexadecimal digits are needed for a 64-bit WEP key? Give any 5 examples.

ANSWER

64-bit WEP key = 40 bits for key + 24 bits IV (Initialization Vector)

40 bits = 10 hexadecimal digits (each hex digit = 4 bits)

Example keys (hexadecimal, 10 digits each):

1A2B3C4D5E

F0E1D2C3B4

A1B2C3D4E5

1234567890

ABCDE12345

f) What command do you type to move from Router> to Router# mode?

From User EXEC mode (Router>) to Privileged EXEC mode (Router#), type:

enable

Q2)B Write a program to display Star Topology.

[20Marks]

ANSWER

```
#include <stdio.h>

int main() {
   int nodes, i;

   printf("Enter the number of nodes in the Star Topology: ");
   scanf("%d", &nodes);

   if(nodes < 2) {
      printf("A star topology requires at least 2 nodes.\n");
      return 1;
   }
```

```
    printf("\nStar Topology Diagram:\n\n");

    // Print central hub
    printf("       [Hub]\n");

    // Print connections to nodes
    for(i = 1; i <= nodes; i++) {
        printf("        |\n");
        printf("       Node%d\n", i);
    }

    printf("\nNote: All nodes are connected to the central hub.\n");

    return 0;
}
```

Q3. Viva

[5Marks]

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Q1) A Write the answer of the following question (ANY 5) [10Marks]

b) Explain steps to identify a phishing email.

ANSWER

Steps to identify a phishing email:

Check the sender's email address – It may look similar but contain small spelling changes.

Look for urgent language – Messages like "Act immediately" or "Your account will be suspended".

Inspect links carefully – Hover over links to see if they lead to suspicious websites.

Check for spelling and grammar mistakes – Many phishing emails contain errors.

Unexpected attachments – Attachments may contain malware.

Requests for sensitive information – Legitimate organizations rarely ask for passwords or banking details via email.

c) What is the function of a switch in a computer network?

ANSWER

A network switch connects multiple devices within a Local Area Network (LAN) and forwards data to the correct device using MAC addresses.

Functions:

Connects computers, printers, and servers in a LAN.

Uses MAC address table to send data only to the intended device.

Reduces network collisions compared to hubs.

Improves network efficiency and performance.

d) How many hexadecimal digits are needed for a 64-bit WEP key? Give any 5 examples.

ANSWER

A 64-bit WEP key uses 40 bits for the key + 24 bits for the initialization vector.

Since 1 hexadecimal digit = 4 bits, a 40-bit key requires:

10 hexadecimal digits

Examples:

1A2B3C4D5E

F1E2D3C4B5

ABCDE12345

1234567890

A1B2C3D4E5

g) Create a dummy phishing email and explain indicators of phishing.
Example Dummy Phishing Email

ANSWER

Subject: URGENT – Your Bank Account Will Be Suspended

Dear Customer,

We detected unusual activity on your bank account.
Your account will be suspended within 24 hours unless you verify your details.

Please click the link below immediately to confirm your account information:

http://secure-bank-verification-login.com

Failure to verify your account will result in permanent suspension.

Thank you,
Bank Security Team

f) What is the primary purpose of a banner MOTD? How can it serve both a legal and security function?
ANSWER

Banner MOTD (Message of the Day) displays a message when a user logs into a network device such as a router or switch.

Example command (Cisco device):

Router(config)# banner motd #Authorized users only#

Purposes:

Legal Function

Displays a warning that the system is for authorized users only.

Notifies users that activity may be monitored.

Security Function

Discourages unauthorized access attempts.

Provides evidence that intruders were warned before accessing the system.

Q2)B Write a program to displays a Mesh Topology.

[20Marks]

ANSWER

```
#include <stdio.h>

int main() {
   int nodes, i, j;

   printf("Enter the number of nodes in the Mesh Topology: ");
   scanf("%d", &nodes);

   if(nodes < 2) {
      printf("A mesh topology requires at least 2 nodes.\n");
      return 1;
   }

   printf("\nMesh Topology Connections:\n\n");

   // Print all connections between nodes
   for(i = 1; i <= nodes; i++) {
```

```
        for(j = i+1; j <= nodes; j++) {
            printf("Node%d <--> Node%d\n", i, j);
        }
    }

    printf("\nNote: In a mesh topology, each node is connected to every other node.\n");

    return 0;
}
```

Q3. Viva

[5Marks]

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Q1) A Write the answer of the following question (ANY 5) [10Marks]

a) Why is it important to configure an enable (EXEC mode) password on a switch?

ANSWER

Configuring an enable password on a switch is important because it protects access to privileged EXEC mode. Privileged mode allows administrators to perform critical operations such as configuring the device, viewing sensitive information, and managing network settings.

Without an enable password, unauthorized users could enter privileged mode and modify the configuration, which may compromise the network. Therefore, it helps in preventing unauthorized access and improving network security.

b) Explain the role of VLAN1 in a switch's initial configuration

ANSWER

VLAN1 is the default VLAN on most Cisco switches.

All switch ports are assigned to VLAN1 by default until they are configured otherwise.

It provides a default management interface for the switch (e.g., assigning an IP to VLAN1 allows remote management via Telnet or SSH).

However, for security reasons, VLAN1 is often changed or isolated because it's widely known and a common target for attacks.

c) Configure Static NAT on a Router (Cisco Packet Tracer / GNS3)

Static NAT maps one private IP address to one public IP address.

Example

Private IP: 192.168.1.10

Public IP: 200.1.1.10

Command Sequence

enable

configure terminal

interface gigabitEthernet0/0

ip address 192.168.1.1 255.255.255.0

ip nat inside

no shutdown

exit

interface gigabitEthernet0/1

ip address 200.1.1.1 255.255.255.0

ip nat outside

no shutdown

exit

ip nat inside source static 192.168.1.10 200.1.1.10

Explanation

ip nat inside → Marks the internal (private) network interface.

ip nat outside → Marks the external (public) network interface.

ip nat inside source static → Creates a permanent mapping between private and public IP.

f) Define Hashing

ANSWER

Hashing is a process of converting data (such as a password or file) into a fixed-length string of characters using a mathematical algorithm called a hash function.

Key points:

- **It is widely used in data security and password storage.**
- **The output is called a hash value or digest.**
- **Hashing is one-way, meaning the original data cannot easily be retrieved from the hash.**

Example algorithms: MD5, SHA-256.

g) What are Cyber Security Policies

ANSWER

Cybersecurity policies are a set of rules, guidelines, and procedures designed to protect an organization's information systems and data from cyber threats.

They define:

- **How users should access and use systems**
- **How to protect sensitive data**
- **Procedures for incident response and security management**

Examples of policies:

- **Password policy**
- **Data protection policy**
- **Network access policy**
- **Incident response policy**

Q2) B Write a program to verify the configuration using show running- config and ping commands

[20Marks]

ANSWER

```
#include <stdio.h>
#include <stdlib.h>
```

```c
int main() {
    char ip[50];
    char configFile[] = "running-config.txt"; // Simulated running-config file
    FILE *fp;
    char line[200];

    printf("=== Configuration Verification Program ===\n");

    // 1. Display running configuration
    printf("\nReading running configuration from %s...\n", configFile);
    fp = fopen(configFile, "r");
    if(fp == NULL) {
        printf("Error: Cannot open %s\n", configFile);
        return 1;
    }

    printf("\n===== Running Configuration =====\n");
    while(fgets(line, sizeof(line), fp)) {
        printf("%s", line);
    }
    printf("===== End of Running Configuration =====\n\n");
    fclose(fp);

    // 2. Ping a host to verify connectivity
    printf("Enter IP address to ping: ");
    scanf("%s", ip);

    printf("\nPinging %s...\n", ip);

    // Execute ping command (Linux: ping -c 4, Windows: ping)
    char command[100];
#ifdef _WIN32
    snprintf(command, sizeof(command), "ping %s", ip);
#else
    snprintf(command, sizeof(command), "ping -c 4 %s", ip);
#endif

    system(command);

    return 0;
}
```

Q3. Viva

[5Marks]

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Q1) A Write the answer of the following question (ANY 5) [10Marks]

a) List and explain all five IP address classes with examples

ANSWER

IP addresses are divided into classes A–E based on the first octet. Each class serves a specific purpose:

class	First Octet Range	Network/Host Bits	Purpose	Example
A	1–126	8 network / 24 host	Large networks (few networks, many hosts)	10.0.0.1
B	128–191	16 network / 16 host	Medium-sized networks	172.16.0.1
C	192–223	24 network / 8 host	Small networks	192.168.1.1
D	224–239	Multicast address	Used for multicasting (sending to multiple hosts)	224.0.0.1
E	240–254	Experimental / reserved	Reserved for research and experimental purposes	240.0.0.1

b) Explain the format and example of an IPv6 address

ANSWER

Example format: 2001:0db8:85a3:0000:0000:8a2e:0370:7334

Shortened version: 2001:db8:85a3::8a2e:370:7334

IPv6 provides:

Vast address space (~3.4×10^38 addresses)

Built-in security (IPSec)

Simplified routing and no need for NAT

d) Configure Dynamic NAT with a Pool of Public IP Addresses

ANSWER

Example:
Private Network: 192.168.1.0/24
Public IP Pool: 200.1.1.10 – 200.1.1.20

Command Sequence (Cisco Router)
enable
configure terminal

access-list 1 permit 192.168.1.0 0.0.0.255

ip nat pool PUBLIC_POOL 200.1.1.10 200.1.1.20 netmask 255.255.255.0

ip nat inside source list 1 pool PUBLIC_POOL

interface gigabitEthernet0/0
ip nat inside
exit

interface gigabitEthernet0/1
ip nat outside
exit

Explanation:

access-list 1 permit 192.168.1.0 0.0.0.255 → Defines the private network allowed for NAT.

ip nat pool PUBLIC_POOL → Creates a pool of public IP addresses.

f) Define Cyber Security

ANSWER

Cyber Security is the practice of protecting computers, networks, programs, and data from digital attacks, unauthorized access, and damage.

Goals include Confidentiality, Integrity, and Availability (CIA triad).

Measures include firewalls, antivirus, encryption, access control, and security policies.

g) Explain Ethical Hacking

ANSWER

Ethical Hacking is the practice of legally testing systems and networks for vulnerabilities to prevent malicious attacks.

Ethical hackers (white hats) use the same techniques as attackers but with permission.

Objectives:

Identify security weaknesses

Improve network security

Ensure compliance with regulations

Tools used: Nmap, Metasploit, Wireshark, Burp Suite

Q2) B Write a program to determine IP address class.

[20Marks]

ANSWER

```
def get_ip_class(ip_address):
    # Split the IP address into octets
    octets = ip_address.split(".")

    if len(octets) != 4:
        return "Invalid IP address format"

    try:
        first_octet = int(octets[0])
    except ValueError:
        return "Invalid IP address"

    # Determine IP class based on first octet
    if 1 <= first_octet <= 126:
        return "Class A"
    elif 128 <= first_octet <= 191:
        return "Class B"
    elif 192 <= first_octet <= 223:
        return "Class C"
    elif 224 <= first_octet <= 239:
```

```python
        return "Class D (Multicast)"
    elif 240 <= first_octet <= 254:
        return "Class E (Experimental)"
    else:
        return "Invalid or reserved IP address"

# Example usage
ip = input("Enter an IP address: ")
ip_class = get_ip_class(ip)
print(f"The IP address {ip} belongs to {ip_class}")
```

Q3. Viva

[5Marks]

______________________________Slip 9__________________________________

Savitribai Phule Pune University
S.Y.B.Sc(Computer Science) NEP - 2023
S.Y.B.Sc.(CS) Practical Examination MAR/APRIL – 2025 - 2026
Lab Course SEC-251-CS-P
Computer Networks

Duration: 3 Hours **Maximum Marks: 35**

Q1) A Write the answer of the following question (ANY 5) [10Marks]

a) Function of a switch in a computer network

ANSWER

A network switch connects multiple devices within a LAN (Local Area Network) and forwards data to the correct device using MAC addresses. It receives a data frame and sends it only to the port where the destination device is connected, which improves network efficiency and reduces collisions.

b) Command to assign a static IP in Linux

ANSWER

Using the ip command:

sudo ip addr add 192.168.1.10/24 dev eth0

This assigns the static IP 192.168.1.10 with subnet mask /24 to the interface eth0.

c) What does ip link set eth0 up do?

ANSWER

The command:

ip link set eth0 up

Activates (enables) the network interface eth0.
It brings the interface up, allowing it to send and receive network packets.

f) What does ip route show display?

ANSWER

The command:

ip route show

displays the system's routing table, including:

Default gateway

Network routes

Interface used for each route

Routing metrics

g) Why should two connected systems be in the same subnet?

ANSWER

Two systems must be in the same subnet to communicate directly without a router.
If they are in different subnets:

The packet must go through a router or gateway.

Q2)Write a C program to display a Star Topology.

[20Marks]

ANSWER

```
#include <stdio.h>

int main() {
   printf("      *\n");
   printf("      |\n");
   printf("*------*------*\n");
   printf("      |\n");
   printf("      *\n");

   return 0;
}
```

Q3. Viva

[5Marks]